
IAL Biology Unit 5

Rapid Revision

WBI15 — Respiration, Internal Environment, Coordination and Gene Technology

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* study note
(20 past papers and mark schemes, June 2019 - June 2026)

90 marks · 1 h 45 min · Q1–Q7 = 70 marks · final question = 20 marks on the pre-released article · one 6-mark (*)

A* rule: grade A overall ($\geq 480/600$ UMS) **and** $\geq 90\%$ across Units 4, 5 and 6 combined ($\geq 270/300$). The A* is decided entirely by A2 performance.

TRIAGE — WHAT THIS PAPER REWARDS

WHERE THE MARKS ARE

Nerves and synapses (10) · heart and exercise (10) · kidney (8) · respiration (8) · gene expression (7) · muscle (7). **Nerves, heart, kidney and respiration carry about 36 of the 70 non-article marks.**

WHAT THE MARK SCHEMES REWARD THAT STUDENTS MISS

1. Exact wording — schemes reject "signals", "produce energy", "ATPase", "sodium" unqualified. **2.** Chains in the **right order** — several schemes say "mps need to be in logical order". **3.** Name which groups when commenting on error bars. **4.** Add a methodology comment — it is often the easiest mark. **5.** Level 3 needs the **molecular** mechanism, not the physiology.

UNIT 5 — SHEET 8 · RESPIRATION

Stage	Where	Products
Glycolysis	Cytoplasm	2 pyruvate · net 2 ATP (substrate-level) · 2 reduced NAD. Glucose phosphorylated using 2 ATP → hexose phosphate → 2 triose phosphate
Link reaction	Matrix	Pyruvate oxidised by NAD and decarboxylated → acetyl + coenzyme A → acetyl CoA; CO ₂ ; reduced NAD
Krebs cycle	Matrix	Acetyl (2C) + oxaloacetate (4C) → citrate (6C) → back to oxaloacetate. Per turn: 2 CO₂ · reduced NAD · reduced FAD · 1 ATP
Oxidative phosphorylation	Inner membrane / cristae	Most of the ATP; water

CHEMIOSMOSIS — THE MOST-ASKED CHAIN IN UNIT 5. LEARN IT WORD-PERFECT.

1. Reduced NAD and reduced FAD carry hydrogen to the ETC on the **inner membrane (cristae)** · **2.** hydrogen **splits into protons and electrons** · **3.** electrons pass along the chain, **releasing** energy — "**REJECT produce energy**" · **4.** energy **pumps protons from the matrix into the intermembrane space**, building a gradient · **5.** protons diffuse back through **ATP synthase** — chemiosmosis · **6.** ADP + Pi → ATP · **7.** **oxygen is the final electron acceptor**, forming water. **REJECT "ATPase"** — **write ATP synthase**. "IGNORE ADP + Pi → ATP without reference to ATP synthase." ETC location: **inner membrane / cristae** — "IGNORE membrane or intermembrane space".

Inhibitor logic — **always the same**: block the ETC → oxygen not used → no proton gradient → no oxidative phosphorylation → only glycolysis → lactate → pH falls → enzymes denature. Cyanide blocks cytochrome oxidase; CO binds haem; TTX/amytal act elsewhere.

Anaerobic: ETC stops → reduced NAD not oxidised → pyruvate **reduced to lactate**, **regenerating NAD** so glycolysis continues (2 ATP). **Lactate goes to the liver and is oxidised back to pyruvate** — it is **never** "used in the Krebs cycle" directly.

Oxygen uptake stays high after exercise because oxygen is needed to convert lactate back to pyruvate/glucose, replenish glycogen, re-oxygenate myoglobin and repay the oxygen debt.

Fermentation: pyruvate **decarboxylated to ethanal**, reduced to **ethanol**, regenerating NAD.

RQ — DEFINITION IS WORTH A MARK ON ITS OWN

RQ = volume of CO₂ produced ÷ volume of O₂ consumed (same time). Both the ratio **and** "volume produced / volume used" are needed.

"Ignore amount"; "IGNORE number / value as an alternative to ratio".

Carbohydrate **1.0** · protein **~0.9** · lipid **0.7** · anaerobic carbohydrate **>1.0**. RQ has **no units**.

UNIT 5 — SHEET 9 · MUSCLES

SLIDING FILAMENT — "MPs NEED TO BE IN LOGICAL ORDER TO GAIN CREDIT"

1. Action potential along the sarcolemma and down the **T-tubules** · **2.** **Ca²⁺ released from the sarcoplasmic reticulum** · **3.** Ca²⁺ binds **troponin**, which changes shape · **4.** troponin moves **tropomyosin**, exposing the **myosin-binding sites on actin** · **5.** myosin heads bind → **cross-bridges** · **6.** **ADP and Pi released** → head tilts = the **power stroke**, pulling actin towards the centre · **7.** **ATP binds**, breaking the cross-bridge · **8.** **ATP hydrolysed**, energy returns the head to its original position · **9.** repeats while Ca²⁺ present; Ca²⁺ actively transported back into the SR to relax.

The repeated trap: ATP hydrolysis moves the head **back** — "**Do not accept the power stroke**" for that step. The power stroke is driven by release of ADP and Pi.

Band / line	Contains	On contraction
Z line	Ends of sarcomere; actin attached	Move closer together
I band	Actin only	Shortens
A band	Full length of myosin	Stays the same
H zone	Myosin only	Shortens

Feature	Slow twitch (I)	Fast twitch (II)
Respiration	Aerobic	Anaerobic (glycogen, creatine phosphate)
Mitochondria / myoglobin / capillaries	Many / much / extensive — red	Few / little / sparse — pale
Sarcoplasmic reticulum	Less	Extensive — more Ca ²⁺ released faster
Diameter / contraction	Smaller; slow, sustained, fatigue-resistant	Larger; rapid, powerful, fatigues quickly

Ligament bone→bone, elastic (elastin), **maintains joint stability**. **Tendon** muscle→bone, **inelastic** (collagen), **transmits force to the bones**. Only **cardiac** muscle is myogenic. Antagonistic pairs: flexor bends, extensor straightens; to hold a joint still **both contract**.

UNIT 5 — SHEET 10 · HEART, VENTILATION & EXERCISE

Myogenic = contraction is initiated **within the cardiac muscle itself, without a nerve impulse**.

THE CONDUCTION CHAIN

SAN (right atrium wall) = pacemaker, generates **waves of depolarisation / excitation** — "**IGNORE impulses**" → spreads across both atria → **atrial systole** → **non-conducting tissue** stops it reaching the ventricles directly → **AVN delays** the wave (atria finish emptying) → **bundle of His** in the septum → **Purkyne fibres** → ventricles contract **from the apex upwards**.

ECG: P = atrial depolarisation · QRS = ventricular depolarisation · T = ventricular **repolarisation**. **heart rate = $60 \div \text{time for one cycle (s)}$** · **cardiac output = heart rate × stroke volume**.

When comparing two traces you **must name the PQRST waves** — "unless they refer to labels PQRST they will not get the marks".

CONTROL OF HEART RATE — SECOND MOST-ASKED CHAIN

exercise → more respiration → **more CO₂** → **blood pH falls** → **chemoreceptors** (carotid and aortic bodies, medulla) detect it → impulses to the **cardiovascular control centre in the medulla** → more impulses along **sympathetic (accelerator) neurones to the SAN** → SAN **depolarises more frequently** → heart rate and force of contraction increase. Adrenaline also binds SAN receptors. To slow: **parasympathetic / vagus nerve**.

"Do not accept signals / messages." **"Ignore increased oxygen demand"** as the stimulus — receptors detect **CO₂ / pH**.

Ventilation: at rest, **stretch receptors** in the lungs inhibit the inspiratory centre → muscles relax → exhalation. In exercise: more CO₂ / lower pH → chemoreceptors → respiratory centre in the medulla → more impulses to **intercostal muscles and diaphragm** → rate and depth increase.

Blood distribution in exercise: cardiac output ↑ · flow to skeletal and cardiac muscle ↑ · to the **skin** ↑ (vasodilation, thermoregulation) · to kidneys and abdominal organs ↓ · to the **brain unchanged**.

Trained athletes: larger/stronger cardiac muscle → greater stroke volume → **lower resting heart rate** · more slow-twitch fibres, mitochondria, myoglobin and capillaries · switch to anaerobic later → less lactate / smaller oxygen debt.

UNIT 5 — SHEET 11 · HOMEOSTASIS, HORMONES & KIDNEY

Homeostasis = "maintaining the body in a state of **dynamic equilibrium**". **Negative feedback reverses** the change; **positive feedback amplifies** it. The point most answers omit: **when receptors detect the condition is back to normal, the effector stops being stimulated** — that is what "narrow limits" means.

Thermoregulation: **thermoreceptors** (skin and hypothalamus) → **hypothalamus** → motor/sympathetic neurones → effectors. Too hot: **vasodilation of skin arterioles**, sweating, hair erector muscles relax. Too cold: vasoconstriction, shivering, erector muscles contract. **"Ignore control centre unqualified"** — name the hypothalamus.

	Peptide (insulin, ADH, adrenaline, GH)	Steroid (oestrogen, testosterone)
Solubility	Water-soluble — cannot cross the membrane	Lipid-soluble — diffuses through
Receptor	On the cell surface membrane	Inside the cell — cytoplasm or nucleus
Next step	Activates a second messenger (cAMP) → transcription factors	Forms a hormone-receptor complex which acts as a transcription factor , binding the promoter

THE KIDNEY — THREE CHAINS

Ultrafiltration: afferent arteriole **wider** than efferent → **high hydrostatic pressure** → small molecules forced through **fenestrations, basement membrane and between podocytes**. The **basement membrane** stops plasma proteins. Filtered: water, glucose, amino acids, urea, ions. Not filtered: blood cells, plasma proteins.

Selective reabsorption (PCT): Na⁺ **actively transported** out of tubule cells into the blood → Na⁺ low inside → Na⁺ diffuses in from the lumen through a **co-transport protein carrying glucose** → glucose into the blood by facilitated diffusion. PCT cells have **microvilli** (large SA) and **many mitochondria** (ATP).

Loop of Henle: limbs flow in **opposite directions** · ascending limb **actively transports Na⁺ and Cl⁻ out** ("**do not accept sodium**" — write ions) and is **impermeable to water** → medulla has **low water potential** → water leaves the descending limb by osmosis → effect **multiplied** down the loop → water drawn out of the **collecting duct** → concentrated urine.

ADH: **osmoreceptors in the hypothalamus** detect a fall in blood water potential → ADH **made in the hypothalamus**, released from the **posterior pituitary** → acts on the **collecting duct and DCT** ("ignore loop of Henle") → binds receptors → vesicles of **aquaporins** fuse with the membrane → more water reabsorbed by osmosis → small volume of concentrated urine → negative feedback.

Desert mammals: longer loops / thicker medulla · more Na⁺/Cl⁻ pumps · greater countercurrent effect · more aquaporins · more mitochondria in the thick ascending limb.

UNIT 5 — SHEET 12 · NERVES, SYNAPSES, BRAIN & EYE

RESTING POTENTIAL → ACTION POTENTIAL

Resting (−70 mV): Na^+/K^+ pump moves **3 Na^+ out, 2 K^+ in** using ATP → gradients → voltage-gated Na^+ channels closed, membrane **more permeable to K^+** → K^+ diffuses out → inside negative.
Action potential: stimulus opens some Na^+ channels → **threshold (−55 mV)** → voltage-gated Na^+ channels open (**positive feedback**) → **depolarisation to +40 mV** → Na^+ channels close, **voltage-gated K^+ channels open** → **repolarisation** → K^+ channels slow to close → **hyperpolarisation** → **refractory period** makes impulses discrete and one-way.
All-or-nothing: a stronger stimulus gives **more frequent** impulses, not bigger ones.

SYNAPTIC TRANSMISSION — THE MOST-ASKED NERVOUS CHAIN

action potential reaches the presynaptic knob → **voltage-gated calcium ion channels open, Ca^{2+} diffuse in** → vesicles **move to and fuse with** the presynaptic membrane → neurotransmitter released by **exocytosis** → diffuses across the **synaptic cleft** → binds **complementary receptors** on the postsynaptic membrane → **ligand-gated Na^+ channels open** → depolarisation → action potential if threshold reached → neurotransmitter **broken down (acetylcholinesterase) or taken back up**.
 "IGNORE sodium ion channel as a receptor." "Do not accept sodium unqualified" — write **sodium ions**.

Myelination — 3 points: myelin is an **insulator**, so ions cannot cross where it is myelinated · action potentials occur **only at the nodes of Ranvier**, where voltage-gated Na^+ channels are concentrated · the impulse **jumps from node to node — saltatory conduction** — so it is faster. "Ignore myelinated neurone has insulation" — say myelin is an insulator.

Drugs: nicotine mimics ACh (stimulates) · **lidocaine and TTX block voltage-gated Na^+ channels** (no depolarisation) · MDMA increases serotonin release · **SSRIs block reuptake proteins** · **L-DOPA crosses the blood-brain barrier and is converted to dopamine** (dopamine itself cannot cross) · acetylcholinesterase inhibitors keep ACh in the cleft.

Parkinson's: death of dopamine-producing neurones in the **substantia nigra** → less dopamine binds receptors → fewer action potentials → less muscle control.

Imaging	How	Shows
CT	X-rays → cross-sectional image	Structure only; quick, cheap; ionising radiation
MRI	Magnetic field + radio waves	Detailed soft-tissue structure; no radiation; not activity
fMRI	Active areas get more blood / oxyhaemoglobin , which absorbs radio waves differently	Activity in real time
PET	Radioactive tracer (glucose) → active cells take up more → positrons → gamma rays	Metabolic activity

Rods: in the **dark** rhodopsin intact, Na^+ channels **open**, membrane **depolarised**, inhibitory **glutamate released**, bipolar neurone hyperpolarised — no impulse. In the **light** rhodopsin is **bleached to opsin + trans-retinal**, Na^+ channels **close**, membrane **hyperpolarised**, no glutamate, bipolar neurone **depolarised** → impulse.

Pupil reflex: bright light → **circular muscles contract, radial relax** → pupil constricts. Dim light / adrenaline → radial contract, circular relax. **"DO NOT ACCEPT ciliary muscles."**

Habituation = a decreased response to a repeated, prolonged, **non-threatening** stimulus. Mechanism: fewer **Ca^{2+} channels** open → less neurotransmitter → less depolarisation.

UNIT 5 — SHEET 13 · PLANTS & GENE EXPRESSION

	Pr (P660)	Pfr (P730)
Absorbs	Red , 660 nm	Far-red , 730 nm
Converted to	Pfr (rapidly)	Pr (rapidly); slowly in the dark
Activity	Inactive	Physiologically active — promotes germination

The rule: the last flash wins. Last treatment red → Pfr high → germination. Last far-red → Pfr converted back to Pr → no germination.
THE SINGLE MOST IMPORTANT REJECTION IN THIS TOPIC: "DO NOT ACCEPT Pfr acts as a transcription factor." Pfr **activates** transcription factors; it is not one. Chain: Pfr enters the nucleus → activates a transcription factor → transcription of germination genes → gibberellin → amylase → starch broken down.

Auxin / phototropism: made in the shoot tip → **accumulates on the shaded side** → stimulates **cell elongation** there → shoot grows **towards the light** ("do not accept bend towards light"). Mechanism: **H^+ pumped into the cell wall** → **pH falls** → **hydrogen bonds between cellulose microfibrils broken** → **water enters by osmosis** → **cell elongates**.

Gibberellin germination: seed absorbs water → embryo releases gibberellin → acts via **transcription factors** in aleurone cells → **amylase gene switched on** → amylase hydrolyses **starch to maltose/glucose** → respired for ATP for growth. "IGNORE produces energy".

Transcription factors are proteins that bind the **promoter** so RNA polymerase can bind. **Epigenetics: DNA methylation** (cytosine) silences genes; **histone acetylation** loosens DNA so transcription is possible; deacetylation/methylation tightens it. **The base sequence is unchanged.**

Alternative splicing: gene → pre-mRNA → **spliceosome removes introns, exons joined in different combinations** → different mRNAs → proteins with different primary structures. "Do not accept DNA splicing."

Differentiation: a stimulus → **differential gene expression** (some genes on, some off) → transcription and translation → proteins determining structure and function.

RECOMBINANT DNA — THE 4-MARK CHAIN

1. Isolate the gene with a **restriction endonuclease** (or reverse transcriptase from mRNA → cDNA) · **2.** cut the **plasmid with the same enzyme** → complementary **sticky ends** · **3.** join with **DNA ligase** (phosphodiester bonds) → recombinant DNA · **4.** insert into the host (bacteria: **heat shock / calcium chloride**; plants: *Agrobacterium*, gene gun) · **5.** select transformed cells with marker genes and culture.
 "IGNORE restricting enzyme" · "ignore gene gun" for bacteria · it is the **plasmid** that is inserted, not the gene directly.
Microarray route: extract mRNA → **reverse transcriptase** → cDNA → fluorescent label → hybridise to the array → **fluorescing spots show active genes** → **bioinformatics**. "Do not pick and mix" with the PCR route.

UNIT 5 — SHEET 14 · THE 20-MARK ARTICLE QUESTION

22% of the paper, 6–10 parts of 1–5 marks, each tied to a numbered paragraph. **Budget 25 minutes.**

Type of sub-question	Share	What it is really asking
Unit 5 chain in a new context	~50%	Chemiosmosis, heart-rate control, synapse, kidney, sliding filament — dressed in the article's subject
Synoptic Units 1–4 chain	~30%	Immune response · inflammation · protein structure · natural selection · osmosis · enzymes
Design a method		Antimicrobial test, habituation assay, ECG study, toxicity assay
"State what is meant by..."		A term from the article — ischaemia, sarcopenia, LD ₅₀ , monoclonal antibody
Small calculation / science in society		Ratio, % change, why a placebo, why references are numbered

HOW TO PREPARE THE ARTICLE BEFORE THE EXAM

1. Read it three times; **number every paragraph**. **2.** Write in the margin which Unit 5 or Unit 1–4 topic each paragraph connects to. **3.** Write the mark-scheme chain next to every named process. **4.** Look up every unfamiliar term — expect a "state what is meant by" on one. **5.** For every experiment, ask: IV, controls, error bars, what the control would be. **6.** For every figure, practise one calculation.

The synoptic chains it keeps using:

Immune response — antigen engulfed by macrophage/APC → presented to **T helper** cells → cytokines activate B cells → B cells differentiate into **plasma cells** releasing antibodies → memory cells.

Inflammation — injury → **histamine / cytokines** → vasodilation, increased blood flow → **capillary walls** more permeable ("IGNORE blood vessels / arterioles") → fluid, white cells and antibodies reach the tissue.

Natural selection — random mutation → new allele → selection pressure → survivors reproduce → **allele frequency increases over generations**.

Protein solubility — primary structure → folding → **hydrophilic R groups outside = soluble**; misfolding puts hydrophobic groups outside → aggregates.

THE REJECTIONS BANK — THE MARKS MOST OFTEN LOST

Do not write	Write instead	
	"Signals" / "messages"	Impulses ; or waves of depolarisation for the SAN
	"Produce energy"	Energy is released
	"ATPase"	ATP synthase
	"Sodium" unqualified	Sodium ions
	ATP hydrolysis drives "the power stroke"	It returns the head; the power stroke is release of ADP and Pi
	"Pfr acts as a transcription factor"	Pfr activates transcription factors
	Lactate "used in the Krebs cycle"	Converted to pyruvate in the liver first
	"Control centre" unqualified	Name it — hypothalamus / medulla / CVC

WALKING INTO THE EXAM

Unit 5 — protect 25 minutes for the article; write chains in the right order; name the enzyme, the ion, the structure; hedge every significance claim and add a methodology comment.